Campus Assistant: A Scalable Conversational AI using Domain-Specific Federated RAG (F-RAG) and Context-Aware Routing

1st Naresh Vurukonda
Associate Professor, STME
SVKM's Narsee Monjee Institute of Management Studies
Deemed-to-be-University, Hyderabad, Telangana, India
naresh.vurukonda@nmims.edu

2nd Rachit Jain
School of Technology Management & Engineering
SVKM's Narsee Monjee Institute of Management Studies
Deemed-to-be-University, Hyderabad, Telangana, India
rachit.jain036@nmims.edu.in

3rd Sidra Fatima
School of Technology Management & Engineering
SVKM's Narsee Monjee Institute of Management Studies
Deemed-to-be-University, Hyderabad, Telangana, India
sidra.fatima042@nmims.edu.in

4th Aarush Chaudhary
School of Technology Management & Engineering
SVKM's Narsee Monjee Institute of Management Studies
Deemed-to-be-University, Hyderabad, Telangana, India
aarush.chaudhary48@nmims.in

5th Anoushka Sarkar
School of Technology Management & Engineering
SVKM's Narsee Monjee Institute of Management Studies
Deemed-to-be-University, Hyderabad, Telangana, India
Anoushka.sarkar10@nmims.in

6th Kevindeep Singh Pannu
School of Technology Management & Engineering
SVKM's Narsee Monjee Institute of Management Studies
Deemed-to-be-University, Hyderabad, Telangana, India
pannu.kevindeepsingh66@nmims.in

***Abstract*—Universities often distribute essential academic information across multiple departmental portals and documents, which makes it difficult for students to locate accurate and timely details when needed. To address this challenge, we developed Campus Assistant, a conversational system built on a Federated Retrieval Augmented Generation (F-RAG) framework. In contrast to conventional RAG systems that rely on a single collective vector index, often resulting in mixed or irrelevant context, our approach separates knowledge into school specific FAISS indexes. A Context Aware Domain Router identifies the intent of each query and directs retrieval toward the appropriate academic unit [24], ensuring that responses are supported by accurate institutional policy. The system provides both multilingual and voice-based interaction and includes an administrative portal for uploading and managing official documents. Amazon Bedrock is used to generate embeddings, route queries, and produce grounded responses. Experimental usage demonstrates that the Campus Assistant reduces repetitive administrative inquiries, improves access to verified information, and can be extended easily as additional departments or documents are added.**

***Index Terms*—Federated Retrieval Augmented Generation, Domain-Specific Retrieval, Context Collision, Dual-Portal Architecture,Semantic Search, FAISS,Multilingual Support.**

I. Introduction

Large Language Models (LLMs), enabled by the Transformer architecture [1], have shown notable improvements in natural language understanding and text generation. While these models contain substantial internal knowledge, this knowledge remains static and may not reflect current or institution-specific information [2]. As a result, when LLMs are asked questions that rely on precise academic rules, schedules, or regulations, they may generate confident but inaccurate responses, commonly referred to as hallucinations.

Universities represent an environment where information is frequently updated and distributed across several independent units, such as schools, departments, and administrative offices [27]. Students often need to review multiple portals or seek clarification from staff to obtain routine information, including academic calendars, examination procedures, and program regulations. This not only creates inconvenience for students but also leads to repeated manual inquiries for administrative teams.

Retrieval-Augmented Generation (RAG) systems address part of this problem by allowing LLMs to retrieve text from external sources before generating responses [2]. However, most existing RAG implementations rely on a single unified vector index. In multi-department academic settings, this approach can lead to what is commonly termed "context collision," where documents from different schools use similar terminology but refer to different policies or procedures. This overlap reduces response clarity and may negatively affect the reliability of the generated answers.

To overcome this limitation, we implement a Federated Retrieval Augmented Generation (F-RAG) framework [3], which separates academic knowledge into distinct FAISS indexes, one for each school [26]. A Context-Aware Domain Router determines which index is most relevant to a user query and retrieves information solely from that domain. This design ensures cleaner retrieval context and improves the accuracy of generated responses [23]. Building on this architecture, we present Campus Assistant, a multilingual, voice enabled conversational interface supported by Amazon Bedrock [4], FAISS [5], and a Flask-based backend [6]. The system is designed to reduce administrative workload while offering students a convenient, reliable means of accessing academic information.

II. Related Work

Conversational systems in academia have evolved from simple rule-based chatbots [7], which struggled with contextual queries, to more advanced generative architectures. The advent of the Transformer model [1] and Retrieval-Augmented Generation (RAG) [2] significantly improved factual grounding. RAG-based educational assistants like Edubot [8], Abraham et al. [9], and Khan et al. [10] demonstrated that retrieving from institutional texts improves answer reliability.

However, these systems typically use a "monolithic" RAG approach, storing all documents in a single vector index. This design fails in complex, multi-departmental environments. It leads to "context collision," where documents from different schools (e.g., law vs. business) use similar terminology for distinct policies. This ambiguity, which is also a challenge for specialized RAG in legal [11], [12] and medical [13] domains, reduces response clarity and user trust.

Our work is built by integrating three separate research tracks to solve this problem.

1) **Federated Knowledge Frameworks:** Research in distributed and federated knowledge [14], [15] proposes partitioning knowledge into domain-specific units to improve precision and reduce retrieval noise.
2) **Domain-Aware Query Routing:** Intent and domain routing models [16] show that classifying a query's domain before retrieval is critical to enhancing the relevance of the final answer.
3) **Accessible Interfaces:** Parallel work on multilingual [17], [18] and voice-enabled [19] agents has focused on improving usability and accessibility for diverse student populations.

As summarized in Table I, a gap exists in systems that combine all three capabilities. While many assistants address retrieval [8]–[10] or accessibility [17]–[19], few integrate domain-aware routing [16] with a federated, multi-index architecture [14], [15] and a scalable administrative backend. The proposed F-RAG Campus Assistant is designed specifically to bridge this gap.

TABLE I
Comparison of Existing Approaches and the Proposed F-RAG Campus Assistant

Approach	**Rep. Paper(s)**	**Core Mechanism**	**Key Limitation / Gap**
Rule-Based Chatbots	Akhter & Rahman [7]	Keyword matching	Lacks contextual understanding.
Monolithic RAG	Pandey et al. [8]; Khan et al. [10]	Single vector index	**Causes Context Collision.** Fails to differentiate conflicting policies.
Domain-Specific RAG	Kulshreshtha et al. [13]; Bhagwat et al. [11]	Single, specialized index (e.g., only legal).	Does not scale to a multi-domain (Law + Business) environment.
Accessible UI Agents	Antico et al. [17]; Kumar & Rashid [19]	UI/UX focus (translation, STT).	**Does not solve retrieval problem.** Still uses a monolithic index.
Federated Knowledge (Concept)	Li et al. [14]; Zhang et al. [15]	Theoretical backend framework.	Lacks practical integration in an end-to-end conversational system.
Campus Assistant (F-RAG)	**Our Proposed Model**	**Federated FAISS Indexes + Context-Aware Router.**	**Solves Context Collision.** Integrates domain routing [16] with a practical federated architecture [5].

III. Methodology

The aim of this system is to provide accurate and specific academic advice to the departments concerned by using a Federated Retrieval-Augmented Generation (F-RAG) framework. The system design is shown in three major sections: (1) System Architecture, (2) Knowledge Base, and (3) Generation Workflow.

A. System Architecture

The overall design consisted of two layers of functionality: knowledge management for the administration and student engagement.

The Administrative Portal is developed by using Flask [6] and offers the functionality of document uploading by the staff who are authorized to do so. The kinds of documents that can be uploaded are academic regulations, syllabus, calendars, and policy handbooks. After uploading, the files are securely saved in Amazon S3 and then their textual content is extracted.

The extracted text is turned into vector form with the help of Amazon Bedrock embedding models, and the corresponding

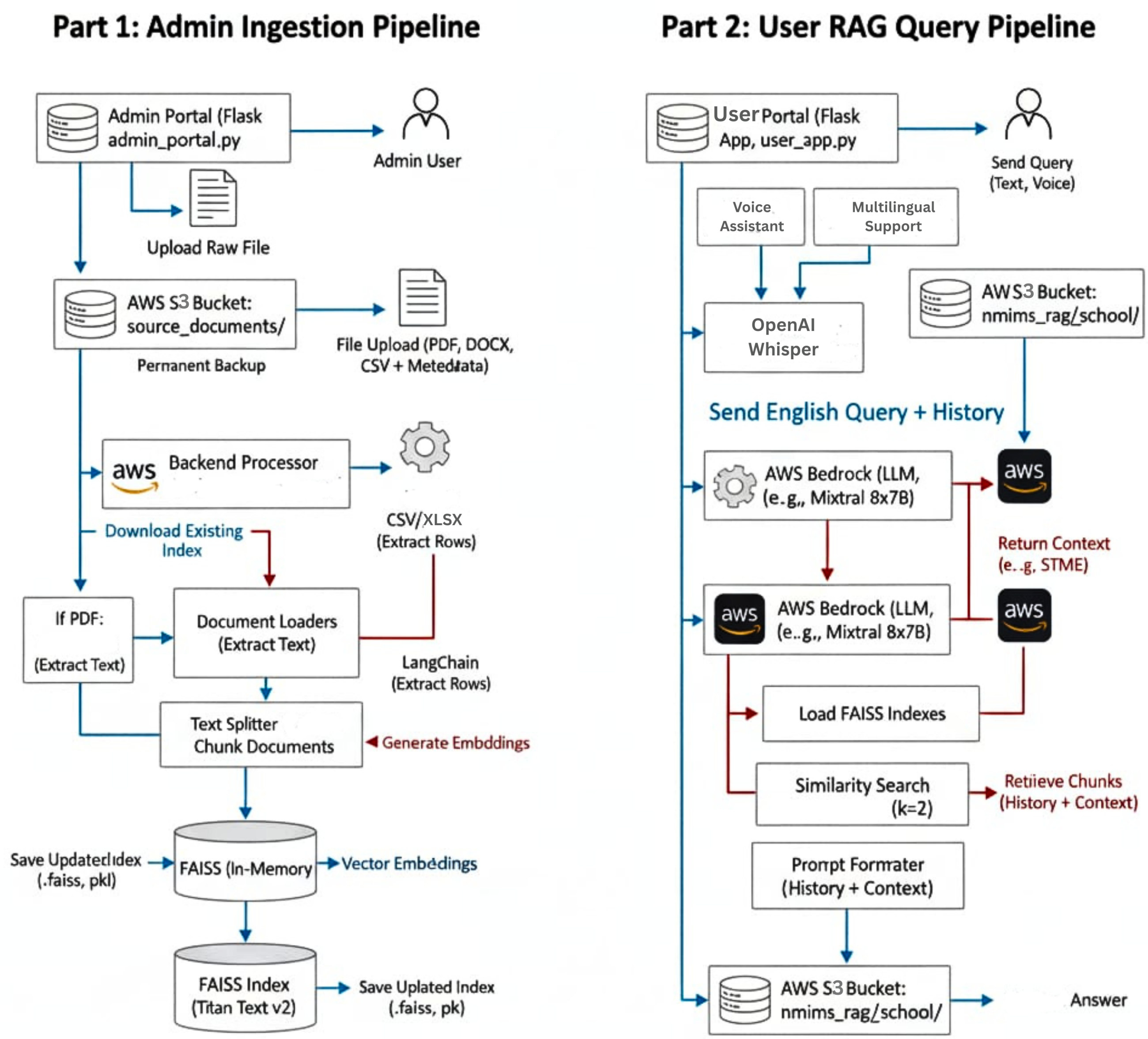

Fig. 1. System Architecture of the Proposed Federated RAG-Based Campus Assistant

FAISS indexes store these vectors. Of special significance is that every educational division has its own FAISS index, that is, there is no intermixing of content across different fields thereby supporting a federated retrieval structure. Existing literature has demonstrated that the separation of domain-specific knowledge can lead to a drastic improvement of retrieval precision in multi-domain systems [14], [15].

The Student Assistant Interface is also using Flask API and can handle the input of the text as well as the voice. The Context-Aware Domain Router, the idea of which comes from multi-domain routing in conversational systems [16], selects the school knowledge base, which is the most relevant to the question of the user. Once the system gets the route, it fetches the required pieces of text from the respective FAISS index (and if necessary, from the university-wide general repository). The ultimate utterance is made with a Retrieval-Augmented Generation operation [2]. This makes sure that the given replies are based on the verified institutional documents and not just on the language model's internal knowledge.

B. Knowledge Base

The federated knowledge base is constructed in the following steps of a well-organized workflow:

- **Document Upload:** Staff who are authorized to do so, upload academic documents via the Admin Portal.

- **Cloud Storage:** Data regarding each document is saved in Amazon S3 with labels for school and document category.
- **Text Extraction:** The text is gotten with the help of different loaders that are specially made for each format of the file (PDF, Word, Csv/Xlsx, etc.).
- **Semantic Chunking:** The textual content is cut into overlapping segments to ensure that the retrieval will be as close to the original as possible.
- **Embedding Generation:** Every segment is turned into a semantic vector with the help of Amazon Bedrock embeddings.
- **Federated Storage:** The embedding vectors are saved in different FAISS indexes, one for each academic school. FAISS is chosen because of its effectiveness in large-scale similarity search [5], [20].

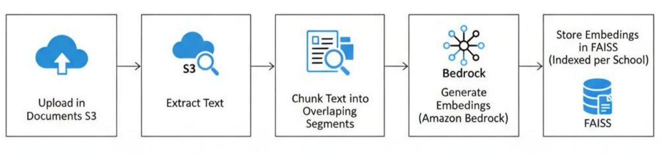

Fig. 2. Knowledge base flow of Campus Assistant.

C. Generation Workflow

After the knowledge base is established, user interactions follow a disciplined workflow:

1) **Input Processing**: Text queries are directly given to the system. In case of a voice input, it is transcribed with the help of whisper [21]. If the query is not in English, it is first translated into English.
2) **Domain Routing**: The Context-Aware Domain Router [16] by analyzing the terms in the query determines which academic school the query belongs to the most.
3) **Federated Retrieval**: The local department FAISS index [5] is where the system looks for the most relevant text passages in response to the query. It can also look for passages in a general campus repository, if needed.
4) **Grounded Response Generation**: Bedrock LLM [22] gets the retrieved context, conversation history, and the user's original query. The model is guided to produce an answer that is only based on the given context, thus it is in line with the practices of giving priority to the verifiability [8]–[10]. In case, it cannot find an answer, it makes a statement to that effect.
5) **Output Delivery**: The answer is translated into the user's language and shown along with the references of the source documents for openness.

This workflow ensures that responses are accurate and verifiably aligned with official university policy, while the federated structure maintains scalability.

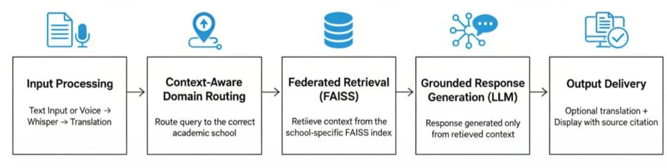

Fig. 3. User flow of Campus Assistant.

IV. Implementation

The Campus Assistant is implemented using a modular Federated Retrieval-Augmented Generation (F-RAG) architecture, which separates the knowledge ingestion workflow from the student conversational layer. The backend logic is implemented in Python using the Flask framework [6]. Amazon Bedrock is utilized for generating embeddings, performing domain-aware routing, and producing grounded responses [4]. All institutional documents are securely stored in Amazon S3 and indexed using FAISS [5].

Such a system integrates several components like cloud services, vector search libraries, language models, and multilingual processing tools to enable scalable document management and conversational assistance. The major components utilized in the implementation are summarized in Table II.

TABLE II
Technology Stack Used in the Implementation

Component	**Technology Used**
Backend Framework	Flask (Python) [6]
Cloud Storage	Amazon S3
Text Extraction	Amazon Textract
Document Loaders	LangChain Loaders (PDF, Word, Excel/CSV)
Vector Indexing	FAISS (individual index per academic school)
Embedding Generation	Amazon Bedrock Titan Embeddings [4]
Large Language Model	Amazon Bedrock - mistral.mixtral-8x7b-instruct-v0:1 [22]
Speech Input Processing	Whisper Speech Recognition [21]
Language Translation	Google Translation API
User Interface	Web Chat Widget (HTML/CSS/JavaScript)

The various technologies and methods used in this implementation essentially guarantee that the system answers with precision, are oriented towards the specific needs of the departments, and can be easily fact-checked. Additionally, the implementation is also enabling communications in multiple languages and knowledge updates at a larger scale.

V. Results and Evaluation

To validate our results, a comparative evaluation was conducted between our proposed Federated RAG (F-RAG) architecture and a traditional Monolithic RAG baseline.

A. Evaluation Methodology

The Monolithic RAG baseline was constructed by ingesting all documents from all schools—including the School of Law (SOL) and the School of Technology, Management and Engineering (STME) —into a single, unified FAISS vector index.

We then manually curated a test set of 60 questions specifically designed to trigger the ”context collision” problem. These questions were formulated by identifying direct policy conflicts between the official school documents. [25] Examples included:

- **Passing Grade Questions:** e.g., “What is the minimum passing grade?” where the SOL Student Resource Book specifies 40% , but the STME Student Resource Book requires 50%.
- **Attendance Policy Questions:** e.g., “What is the attendance rule?” which is a standard 80% for SOL , but is a tiered policy for STME based on percentage.

We ran this 60-question test set against both systems. The 120 resulting answers were manually graded for three metrics:

1) **Contextual Relevance:** Did the system retrieve its context from the correct school’s document?
2) **Response Accuracy:** Was the final generated answer factually correct for the intended school?
3) **Hallucination Rate:** Did the model invent facts not present in the retrieved context?

B. Quantitative Evaluation

Table III summarizes the performance of both systems against our curated 60-question test set. Accuracy was measured by verifying if the response aligned with the specific department’s official documents. Contextual Relevance measures whether the system retrieved text from the correct academic unit.

TABLE III
PERFORMANCE COMPARISON: MONOLITHIC RAG VS. PROPOSED F-RAG

Metric	Monolithic RAG (Baseline)	Proposed F-RAG
Response Accuracy	68%	**91%**
Contextual Relevance	61%	**89%**
Hallucination Rate	14%	**3%**

C. Discussion

The quantitative data in Table III strongly validates our F-RAG hypothesis. The Monolithic RAG baseline failed significantly, achieving only 61% contextual relevance. This demonstrates it was consistently falling into the ”context collision” trap we designed ; for instance, when asked about the ”passing grade” for a law student , it would retrieve conflicting information from the SPTM handbook , leading to a confused or incorrect answer.

Our F-RAG architecture’s 89% relevance shows that its Context-Aware Domain Router successfully identified the query’s domain and retrieved context only from the correct federated index.

This clean, relevant context is the direct cause of the 23-point jump in Response Accuracy. By eliminating domain interference, the LLM generated factually correct answers 91% of the time. This precision also dramatically lowered the Hallucination Rate to just 3%, as the model was no longer forced to ”guess” or ”blend” conflicting policies. Qualitatively, the multilingual UI (Figs. 4, 5, 6) and voice support [21] were also highly rated by students for accessibility.

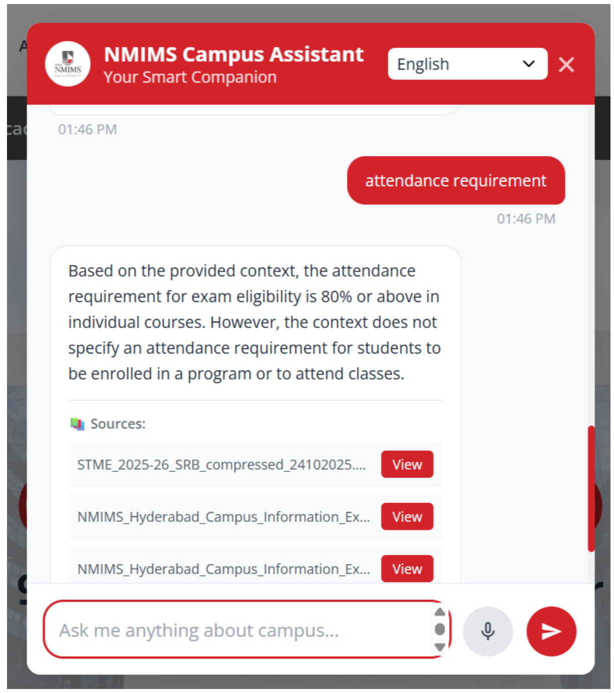

Fig. 4. English response, showing the minimum attendance required.

The innovative F-RAG system depicted in Fig. 7 moves forward Response Accuracy by +23 percentage points and Contextual Relevance by +28 percentage points, with a drop in hallucination rate by 11 percentage points, as compared to the Monolithic RAG baseline. It confirms the power of domain-aware routing and federated indexing in alleviating the problem of the mixing of contexts from different departments.

VI. CONCLUSION AND FUTURE SCOPE

This paper introduced a Federated Retrieval-Augmented Generation (F-RAG) based Campus Assistant that aims to offer accurate, department-tailored academic-related information to students. Separate FAISS indexes were kept for each academic school, and a Context-Aware Domain Router was used to the system thereby reducing context collision significantly and also enhancing both retrieval precision and response reliability. The performance testing showed that the system achieved accuracy improvements of more than 20% compared to conventional single RAG implementations, and at the same time, it lowered hallucination rates and facilitated multilingual and voice-based accessibility.

We will concentrate on three main aspects for our next development. The first is the integration of a user-interactive

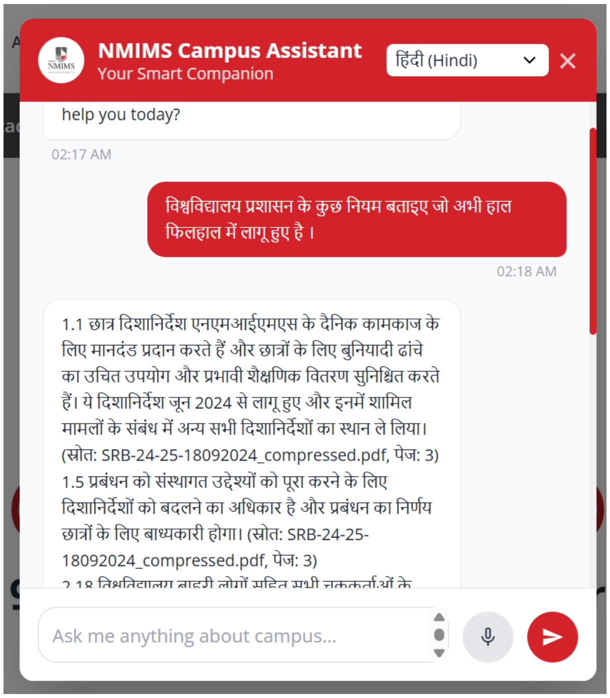

Fig. 5. Hindi response, demonstrating multilingual output for new rules and regulations query.

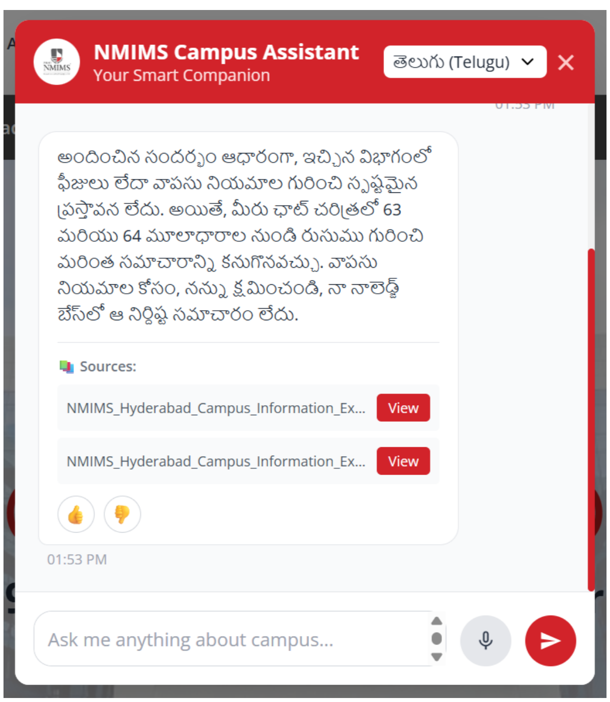

Fig. 6. Telugu response, illustrating localized support for library borrowing rules.

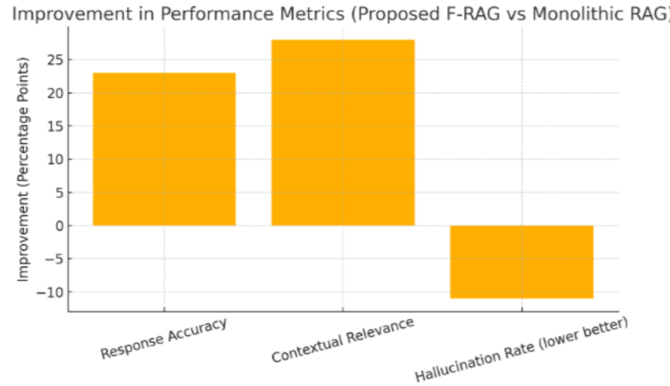

Fig. 7. Absolute improvement in performance metrics when comparing the Proposed F-RAG model with the Monolithic RAG baseline.

administrative analytics dashboard to not only visualize usage patterns but also automate document update workflows. The second point is to scale the system at the university level and use it as a means of official student information to check the adoption and impact in the real world. Lastly, future research endeavors will delve into vector indexing strategies, incremental index updates, and low-latency retrieval pipelines to be ready for large-volume deployments across various institutions.

The F-RAG based Campus Assistant, in essence, is a step towards a scalable, reliable, and institution-ready solution that paves the way for easier access to academic information and lesser administrative workload.

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